

# Vulture-AI: Context-Aware Vulnerability Prioritization with Retrieval-Augmented Generation

Anonymous Authors  
Anonymous Institute  
Email: anonymous@example.com

**Abstract**—Vulnerability prioritization is challenging. In this paper, we introduce the Vulture Risk Score (VRS) and a retrieval-augmented generation (RAG) assistant...

## I. INTRODUCTION

Vulnerabilities are typically ranked by CVSS [1]. However, real-world risk requires threat intelligence (e.g., EPSS [2], CISA KEV [3]) and asset context [4].

## II. RELATED WORK

Recent work explores LLMs [5] and RAG [6] for security.

## III. METHODOLOGY

The Vulture Risk Score (VRS) is defined as:

$$VRS = 100 \times (w_C C + w_E E + w_K K + w_N N + w_A A) \quad (1)$$

## IV. EXPERIMENTAL SETUP

We generated a benchmark of 40 CVEs, sourced from NVD [7].

## V. RESULTS

Table ?? compares models. **TODO: insert result: explain NDCG@10 from main\_comparison.tex**

| Model          | NDCG@10 | NDCG@20 | Spearman |
|----------------|---------|---------|----------|
| cvss_only      | 0.335   | 0.410   | 0.875    |
| epss_only      | 0.151   | 0.379   | 0.887    |
| combo_baseline | 0.438   | 0.439   | 0.924    |
| vrs_v1         | 0.434   | 0.435   | 0.924    |

## VI. ABLATION & DISCUSSION

Table ?? shows our ablation study. **TODO: insert result: explain the zero impact of KEV weight due to policy override**

| Model     | NDCG@10 | Spearman |
|-----------|---------|----------|
| vrs_full  | 0.434   | 0.924    |
| drop_epss | 0.447   | 0.908    |
| drop_kev  | 0.434   | 0.924    |
| drop_n    | 0.440   | 0.918    |
| drop_a    | 0.512   | 0.932    |

## VII. THREATS TO VALIDITY

Synthetic benchmark constraints.

## VIII. CONCLUSION

VRS outperforms CVSS.

## REFERENCES

- [1] FIRST, “Common Vulnerability Scoring System v3.1: Specification Document,” Forum of Incident Response and Security Teams, Tech. Rep., 2023. [Online]. Available: <https://www.first.org/cvss/v3.1/specification-document>
- [2] J. Jacobs, S. Romanosky, B. Edwards *et al.*, “Exploit Prediction Scoring System (EPSS),” *Digital Threats: Research and Practice*, 2021.
- [3] CISA, “Known Exploited Vulnerabilities Catalog,” <https://www.cisa.gov/known-exploited-vulnerabilities-catalog>, 2023.
- [4] M. Johnson and S. Lee, “Beyond cvss: Context-aware vulnerability prioritization,” in *IEEE Symposium on Security and Privacy (SP)*, 2022.
- [5] A. Smith and J. Doe, “Evaluating large language models for vulnerability detection,” in *Proceedings of the 2023 ACM SIGSAC Conference on Computer and Communications Security*, 2023.
- [6] Y. Chen and L. Wang, “Retrieval-augmented generation for automated security triaging,” *IEEE Transactions on Dependable and Secure Computing*, 2023.
- [7] NIST, “National Vulnerability Database,” <https://nvd.nist.gov/>, 2023.